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Summary

This paper introduces Adversarial Score Matching (ASM), a novel method for learning unnormalized models, particularly in the presence of missing data. The core idea behind ASM is to update the model parameters by maximizing the score discrepancy, which is the difference between

the scores of the data distribution and the model distribution. This approach contrasts with traditional methods that often rely on auxiliary networks or importance sampling. A key contribution of this work is the introduction of a multilevel Monte Carlo (MLMC) estimator for approximating the score discrepancy, which the authors claim is more computationally efficient than standard importance sampling techniques. The authors also present a theoretical analysis, demonstrating that ASM exhibits mode-seeking behavior, meaning it converges to the modes of the data distribution, and that it has a smaller energy bias compared to Maximum Likelihood Estimation (MLE). The authors provide theoretical results suggesting that ASM provides better imputation performance than MLE. Empirically, the authors evaluate ASM on several unnormalized models and missing data mechanisms, comparing its performance against MLE, a conditional score matching approach, and a recent method called MissDiff. The experimental results, while limited in scope, generally indicate that ASM achieves lower energy values, suggesting better sample quality, and outperforms the baselines in terms of imputation performance. The paper argues that ASM's ability to handle high-dimensional data and its mode-seeking properties make it a valuable tool for learning complex distributions with missing data. Overall, the paper presents a promising new approach to score matching, with a focus on computational efficiency and theoretical properties, although several aspects require further clarification and empirical validation. The use of a multilevel Monte Carlo estimator for score discrepancy is a notable technical innovation, and the theoretical analysis of mode-seeking behavior provides a valuable insight into the method's characteristics. However, the paper's claims about the method's efficiency and its practical advantages over existing methods need more robust empirical support and a more rigorous theoretical foundation.

✓ Strengths

I find several aspects of this paper to be quite compelling. The introduction of Adversarial Score Matching (ASM) as a method for learning unnormalized models with missing data is a significant contribution. The core idea of updating the model by maximizing the score discrepancy is both intuitive and theoretically grounded. Unlike many existing methods, ASM does not require an auxiliary network, which simplifies the learning process and potentially reduces computational overhead. The authors' claim that ASM can be applied to high-dimensional data is also a notable advantage, as many traditional methods struggle in such settings. The use of a multilevel Monte Carlo (MLMC) estimator for the score discrepancy is a technical innovation that addresses the computational challenges associated with estimating this quantity. The authors provide a theoretical argument, in Appendix A.1.5, suggesting that the MLMC estimator has a faster convergence rate than standard importance sampling, which is a promising result. Furthermore, the authors' theoretical analysis of ASM's mode-seeking behavior, presented in Section 5, is a valuable contribution. The idea that ASM converges to the modes of the data distribution, minimizing energy bias, is a desirable property for generative models. The theoretical results, while not fully comprehensive, provide a solid foundation for understanding the method's behavior. The experimental results, while limited in scope, consistently show that ASM achieves lower energy values compared to MLE and other baselines, indicating better sample quality. The authors also demonstrate that ASM performs well in imputation tasks, further highlighting its practical utility. The comparison with MissDiff, a recent method for handling missing data in score-based models, is also a strength, as it provides a relevant benchmark for evaluating ASM's performance. Finally, the paper's focus on addressing the challenges of learning unnormalized models with missing data is highly relevant and timely, as this is a common problem in many real-world applications.

✗ Weaknesses

Despite the strengths of this paper, I have identified several weaknesses that need to be addressed. First, while the authors claim that ASM is a mode-seeking algorithm and has improved mode-seeking behavior compared to MLE, the definition of "mode-seeking" is not rigorously defined within the context of ASM. Although Section 5 attempts to explain this behavior, the explanation is somewhat intuitive and lacks a formal mathematical definition. The connection between minimizing score discrepancy and converging to modes is explained, but could be more rigorous. This lack of a precise definition makes it difficult to fully understand the theoretical underpinnings of the method and its advantages over other approaches. Second, while the authors provide theoretical results suggesting that ASM provides better imputation performance than MLE, the empirical evidence presented in Section 6 is not sufficient to fully support this claim. While the experiments show that ASM achieves lower energy values, which is indicative of better sample quality, the datasets used are not particularly complex, and the experiments lack ablation studies to analyze the impact of different components of the proposed method. This makes it difficult to assess the practical relevance of the theoretical findings. Third, although the authors claim that the multilevel Monte Carlo (MLMC) estimator is more computationally efficient than the traditional importance sampling estimator, the paper does not provide a detailed analysis of the computational complexity of their method. While

Appendix A.1.5 provides a theoretical comparison of convergence rates, a more detailed step-by-step computational complexity analysis is needed to justify the claim of improved efficiency. The paper also does not provide empirical evidence of the computational time savings achieved by the MLMC estimator. Fourth, the paper's writing is not always clear and consistent. As pointed out by one of the reviewers, the notation is inconsistent, particularly in Equations (4) and (5), where $s_{\theta}(x)$ and $s_{\theta t}(x)$ are used without a clear explanation of the difference. This lack of clarity makes it difficult to follow the technical details of the method. Furthermore, the proofs of Theorem 4.2 and Corollary 4.2 are missing from the main body of the paper and the provided appendix, which is a significant oversight. Finally, the theoretical analysis of the proposed method is limited. The convergence rate of $O(T^{-1/2})$ provided in Theorem 4.2 is not exceptionally fast, and the authors do not provide a more in-depth analysis of the convergence properties of the proposed method. The paper also does not provide a theoretical justification for why the proposed method is more efficient than existing methods, beyond the convergence rate analysis of the MLMC estimator. The lack of a detailed analysis of the bias and variance of the estimator used in the proposed method is also a concern. These weaknesses, taken together, significantly impact the overall strength of the paper and need to be addressed to fully validate the authors' claims.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should provide a more rigorous definition of "mode-seeking" within the context of ASM. This could involve a formal mathematical definition, perhaps in terms of the behavior of the score function around modes, and a more detailed explanation of how the score discrepancy is related to the mode-seeking behavior. The authors should also clarify how their definition of mode-seeking differs from other existing definitions in the literature. Second, the authors should provide more empirical evidence to support their theoretical results. This should include experiments on more complex datasets, as well as ablation studies to analyze the impact of different components of the proposed method, such as the MLMC estimator. The authors should also provide a sensitivity analysis of the hyperparameters of the method to ensure that the results are robust. Third, the authors should provide a more detailed analysis of the computational complexity of their method. This should include a step-by-step analysis of the computational cost of the MLMC estimator, as well as a comparison of the computational cost of ASM with other methods, including MLE and other score matching methods. The authors should also provide empirical measurements of the computational time to support their claims of improved efficiency. Fourth, the authors should improve the clarity and consistency of the paper's writing. This includes clarifying the notation, providing a clear explanation of the difference between $s_{\theta}(x)$ and $s_{\theta t}(x)$, and including the proofs of Theorem 4.2 and Corollary 4.2 in the final version of the paper. A table of notations would also be helpful to keep track of the different symbols used throughout the paper. Fifth, the authors should strengthen the theoretical analysis of the proposed method. This could involve providing a more in-depth analysis of the convergence properties of the method, as well as a theoretical justification for why the proposed method is more efficient than existing methods. The authors should also discuss the conditions under which the proposed method is expected to perform well and the conditions under which it might fail. A more detailed analysis of the bias and variance of the estimator used in the proposed method would also be beneficial. Finally, the authors should compare their method to other state-of-the-art methods for learning unnormalized models with missing data. This comparison should include both theoretical and empirical comparisons, and should discuss the advantages and disadvantages of their method compared to other methods, including the computational cost, the imputation performance, and the robustness to different types of missing data mechanisms. These improvements would significantly strengthen the paper and make it a more valuable contribution to the field.

Questions

Based on my analysis, I have several questions that I believe are crucial for a deeper understanding of the proposed method. First, can you provide a more detailed explanation of the mode-seeking behavior of ASM? Specifically, how does the score discrepancy update rule in Equation (6) lead to the method converging to modes of the data distribution? How does your definition of mode-seeking differ from other definitions in the literature, and how does ASM's mode-seeking behavior compare to other methods that also aim to capture the underlying structure of the data? Second, can you provide more empirical evidence to support your theoretical results showing that ASM provides better imputation performance than MLE? Specifically, can you conduct experiments on more complex datasets and include ablation studies to analyze the impact of different components of the proposed method, such as the MLMC estimator? Third, can you provide a more detailed analysis of the computational complexity of your method? Specifically, how does the computational cost of the MLMC estimator scale with the dimensionality of the data and the number of samples? How does the computational cost of ASM compare to other state-of-the-art methods for learning unnormalized models with

missing data, both theoretically and empirically? Fourth, can you clarify the notation used in the paper, particularly the difference between $s_{\theta}(x)$ and $s_{\theta}(t)(x)$? Can you also provide the proofs of Theorem 4.2 and Corollary 4.2? Finally, can you provide a more in-depth analysis of the convergence properties of the proposed method? Specifically, under what conditions is the method expected to converge, and what is the rate of convergence? How does the convergence rate of ASM compare to other methods, and what are the limitations of the theoretical results? These questions are aimed at clarifying the core aspects of the method and its theoretical and practical implications.



Scores

Soundness: 2.5

Presentation: 2.25

Contribution: 2.5

Rating: 4.5



AI Review from ZGCA will be automatically processed



Summary

The paper proposes Adversarial Score Matching (ASM), an adversarial learning algorithm for unnormalized models with missing data that purportedly does not require an auxiliary score network and is applicable to high-dimensional data. The core objective is to minimize the score discrepancy $D_{\text{score}}(p_{\eta} || q_{\theta})$, and the paper introduces a multilevel Monte Carlo (MLMC) estimator to estimate this discrepancy more efficiently than importance sampling. The method is framed as an iterative proximal-gradient-style update (Algorithm 1) and is accompanied by a convergence discussion. The authors additionally present a theoretical perspective suggesting ASM is mode-seeking with smaller energy bias than MLE, and evaluate on toy EBMs, Gaussian graphical models, and a yeast gene expression dataset using the learned model energy as the primary evaluation metric.



Strengths

- Addresses a relevant and challenging problem: learning unnormalized models with missing data, a regime where likelihood-based training is difficult.
- Aspires to eliminate auxiliary score networks, which could simplify training and reduce overhead in high-dimensional settings (Abstract; Section 1).

- Introduces the idea of an MLMC estimator specifically for score discrepancy estimation (Abstract; Section 3/A.1 reference), which, if correctly specified and validated, could be a useful contribution.
- Attempts to connect the objective to mode-seeking behavior, providing a theoretical perspective that could be of independent interest (Sections 5 and 7).

✗ Weaknesses

- Unobservable/ill-defined update: The core estimator (Eq. (5)) and update (Eq. (7)) explicitly require $s_{\eta}(x_i)$, the score of the unknown data distribution, which is not accessible from samples without an auxiliary estimator or an integration-by-parts trick. The paper repeatedly claims to avoid auxiliary networks, but provides no mechanism to obtain s_{η} . This appears to be a fatal flaw in the current formulation.
- Objective inconsistency and conceptual confusion: The paper alternates between minimizing and maximizing the score discrepancy. Section 3 aims to minimize D_{score} ; Section 4.1 (Eq. (6)) defines an argmax over D_{score} and calls the procedure 'adversarial', which is at odds with score matching. The proximal step (Eq. (7)) is not correctly reconciled with either a principled ascent or descent direction under a well-defined surrogate.
- Contradictory claims about sampling: The paper claims 'we do not require samples from the model' (end of Section 4), yet Algorithm 1, line 3, explicitly requires sampling from $q_{\{\theta_t\}}(\cdot)$.
- Convergence analysis lacks rigor and clarity: Assumptions (Section 4.2) are informal or ill-defined (e.g., the $RV(p_1, p_2, p_3)$ notation, 'Bias' definition, and appeal to Hölder continuity). The stated bound is not clearly derived from the algorithm that itself depends on s_{η} and an unspecified unbiased estimator. The role of the proximal regularizer and step-size conditions are not tied to verifiable smoothness properties of the true objective.
- Mode-seeking theory is error-prone and unconvincing: The Gaussian analysis (Section 5) contains numerous notational inconsistencies (e.g., 'diag-bar', mismatched matrix products, unclear limits), and key equalities are not derived. The claimed link between smaller energy bias and practical imputation quality is asserted but not substantiated.
- Experimental design and reporting are insufficient: The main metric is the learned model energy on test samples, not standard imputation metrics (e.g., RMSE/MAE on masked entries, NLL, calibration). Baselines are incomplete given the literature (e.g., sliced score matching variants for scalability; strong conditional diffusion imputation methods). No ablations isolate the contribution of MLMC vs. importance sampling, and there is no runtime/variance analysis to justify efficiency claims. The experiments lack crucial details on architectures, initialization, sampling strategies, and hyperparameters, impeding reproducibility.
- Notation and presentation issues throughout: There are many typos and missing elements (e.g., undefined 'Algorithm 2', incomplete lines 'using e.g.', inconsistent notation for missingness mechanism $p(\cdot|\cdot)$, misleading MCAR description). These issues obscure the core ideas and make verification difficult.

? Questions

- How is $s_{\eta}(x)$ obtained in Eq. (5) and Eq. (7) without an auxiliary score estimator or an integration-by-parts identity? Please provide the exact estimator for s_{η} used in practice, or a derivation showing how it cancels or can be replaced by observable quantities.
- Please reconcile the optimization direction: are you minimizing $D_{\text{score}}(p_{\eta} \parallel q_{\theta})$ (score matching) or maximizing it (Eq. (6))? If the latter, what is the rationale, and how does it lead to model learning rather than divergence from data? Provide a correct derivation of Eq. (7) from the chosen objective.
- You state 'we do not require samples from the model', yet Algorithm 1, line 3 requires sampling from $q_{\{\theta_t\}}$. Which is correct? If sampling is required, how is it performed for unnormalized models that cannot be directly sampled?
- The MLMC estimator: Please include the full estimator (Appendix A.1 is referenced but not present here), its unbiasedness/bias-variance trade-off, level schedule, and complexity analysis compared to importance sampling. Provide empirical variance and wall-clock comparisons under controlled settings.

- Missingness mechanism: The MCAR description and the use of $p(x|x)$ for missingness are unclear. Please formalize the missingness model, the mask distribution, and how conditional scores $s_{\theta}(x|x_{\text{obs}})$ are computed/approximated.
- Experimental metrics: Why is the learned energy f_{θ} on test samples an adequate proxy for imputation quality? Please report standard imputation metrics (e.g., RMSE/MAE on masked entries, NLL/log-likelihood surrogates, calibration) and compare against strong baselines (e.g., conditional diffusion/score models, sliced score matching variants).
- Ablations: Please ablate the MLMC estimator vs. importance sampling, with variance/runtime curves and final performance; ablate the effect of the adversarial/proximal formulation; and test sensitivity to step sizes and sampling budgets.
- Reproducibility: Provide complete architecture details (layers, activations, normalization), optimization hyperparameters, initialization, seeds, sampling methods, and compute budgets. For the gene expression dataset, clarify the exact data source, preprocessing, feature dimensionality, and missingness protocol.

 Limitations

- The current formulation appears to require access to the data score $s_{\theta}(x)$, which is not available in practice without an estimator—undermining the central claim of ‘no auxiliary network’.
- The theoretical mode-seeking analysis is limited to simplistic Gaussian settings and contains notational and derivational errors; it does not convincingly transfer to high-dimensional, nonlinear models.
- Evaluation relies on an internal energy metric rather than standard imputation metrics; the link between ‘lower energy’ and actual imputation quality is unvalidated.
- Lack of ablations and variance/runtime analyses for the proposed MLMC estimator prevents assessing its practical efficiency.
- Potential negative societal impacts were not discussed; while the method is general, applications to sensitive domains (e.g., healthcare) warrant a discussion on imputation biases under MCAR vs. MAR/MNAR and downstream decision risks.

 Image Evaluation

Cross-Modal Consistency: 12/50

Textual Logical Soundness: 8/30

Visual Aesthetics & Clarity: 5/20

Overall Score: 25/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

- Major 1: Tables 1–3 are cited but not present; claims cannot be verified. Evidence: “Table 1 shows the sample quality...”
- Major 2: Algorithm 2 is referenced but missing. Evidence: “(s_t from Algorithm 2).”
- Major 3: Sampling contradiction: text claims no model samples needed, but Algorithm 1 requires sampling from q . Evidence: “we do not require samples from the model.” vs “Sample $\{x_i\}$ from $q_{\theta_t}(\cdot)$...”
- Major 4: Objective sign inconsistency: they aim to minimize D_{score} , yet update maximizes it. Evidence: “we update the model parameter to maximize the score discrepancy”
- Minor 1: Equation numbering/wording inconsistent and duplicated. Evidence: “Equation equation 5...”

- Minor 2: Symbol reuse/mismatch for observed/missing data and mask. Evidence: " $p(x|x)$ " and " $x_i = x_i \odot i + x_i \odot (1-i)$ "

2. Text Logic

- Major 1: Estimator depends on unknown data score $s_\eta(x)$; no practical means given to compute it. Evidence: " $\|s_\eta(x_i) - s_\theta(x_i, x_i)\|^2$ "
- Major 2: Mode-seeking derivations contain incorrect/incomplete Gaussian formulas (e.g., KL, matrix terms) and unclear "diag-bar" setting. Evidence: " $D_{KL}(p_\eta||q_\theta) = 1/2 (\Sigma_\theta^{-1} \Sigma_\eta) - d/2$ "
- Minor 1: Convergence theorem incomplete: theorem not stated; assumptions/notation unclear and unstandard. Evidence: "Theorem shows that the proximal gradient algorithm..."
- Minor 2: Numerous typos obscure meaning (e.g., "denotes(x)", "advice-teacher"). Evidence: "where denotes(x) = (x,x(x))"

3. Figure Quality

- Major 1: No figures/tables included; all quantitative claims uncheckable; blocks result verification. Evidence: "Table 2 shows the sample quality..."
- Minor 1: Algorithm 1 is ambiguous ("using e.g.") and missing referenced components. Evidence: "using e.g." (truncated)

Key strengths:

- Addresses learning unnormalized models with missing data, a relevant and challenging setting.
- Seeks to avoid auxiliary score networks and claims MLMC estimator for efficiency.
- Attempts to analyze mode-seeking behavior.

Key weaknesses:

- Core contradictions (maximize vs minimize D_{score} ; sampling required vs claimed unnecessary).
- Estimator relies on unavailable s_η , undermining practicality.
- Missing tables/algorithms prevent verification of empirical claims.
- Mathematical derivations and notation contain errors and ambiguities.
- Convergence result/theory not properly stated; appendix (MLMC) absent.

Recommendations:

- Clarify objective (minimize D_{score}) and fix Algorithm 1 accordingly.
- Provide a computable estimator that does not require s_η , or specify how s_η is obtained.
- Include all tables/algorithms/appendices; ensure consistent notation for observed/missing variables and mask.
- Correct Gaussian derivations and provide precise theorems with assumptions and proofs.
- Resolve sampling claims; specify exact sampling or avoidance mechanism.



Scores

Originality: **2**

Quality: **1**

Clarity: 1

Significance: 2

Soundness: 1

Presentation: 1

Contribution: 2

Rating: 2

 AI Review from SafeReviewer will be automatically processed **Summary**

This paper introduces Adversarial Score Matching (ASM), a novel approach for learning unnormalized models, particularly in scenarios with missing data. The core idea of ASM revolves around directly estimating model parameters by maximizing the score discrepancy, thereby circumventing the need for auxiliary networks often employed in traditional score matching methods. The authors propose an efficient multilevel Monte Carlo (MLMC) estimator to compute this score discrepancy, which they argue is more computationally efficient than standard importance sampling techniques. The method is presented as a mode-seeking algorithm, which is beneficial for capturing the primary modes of the data distribution, and is evaluated on both synthetic and real-world datasets. The empirical results demonstrate that ASM achieves comparable or better performance than existing methods like MissDiff, especially in high-dimensional settings and with complex missing data patterns. The authors also provide theoretical analysis, including convergence properties and mode-seeking behavior, to support their claims. Overall, the paper presents a promising alternative to existing score matching methods, particularly for high-dimensional data with missing values, by offering a computationally efficient and effective approach for learning unnormalized models.

 Strengths

I find several aspects of this paper to be quite compelling. The introduction of Adversarial Score Matching (ASM) as a method for learning unnormalized models without relying on auxiliary networks is a significant contribution. This approach directly addresses a key limitation of many existing score matching techniques, which often require additional networks to estimate the score function. By directly maximizing the score discrepancy, ASM offers a more streamlined and potentially more efficient approach. Furthermore, the development of a multilevel Monte Carlo (MLMC) estimator for the score discrepancy is a notable technical innovation. The authors convincingly argue that this estimator is more computationally efficient than traditional importance sampling methods, which is crucial for handling high-dimensional data. The paper also

provides a theoretical foundation for ASM, including convergence analysis and a discussion of its mode-seeking behavior. This theoretical analysis, while not without its limitations, provides valuable insights into the properties of the proposed method. Finally, the empirical results presented in the paper, while not exhaustive, demonstrate the potential of ASM to achieve comparable or better performance than existing methods, particularly in challenging scenarios with high-dimensional data and complex missing data patterns. The application of ASM to real-world datasets, such as spliced gene expression data, further highlights its practical relevance and potential impact.

Weaknesses

Despite the strengths of this paper, I have identified several weaknesses that warrant careful consideration. First, the paper's discussion of its relationship to existing methods, particularly those related to adversarial learning and score matching, is not sufficiently detailed. While the authors mention that ASM does not require an auxiliary network, they do not fully explain the nuances that differentiate it from other adversarial score matching techniques. For instance, the paper does not clearly articulate how the direct parameter update in ASM avoids the instability often associated with adversarial training of auxiliary networks. This lack of clarity makes it difficult to fully appreciate the novelty and advantages of ASM. Furthermore, the paper's presentation of the MLMC estimator lacks sufficient detail. While the authors claim it is more computationally efficient than importance sampling, they do not provide a thorough explanation of the variance reduction techniques employed or how these techniques are specifically applied to the score discrepancy estimation. The paper also does not provide a clear derivation of the computational complexity of the MLMC estimator, nor does it compare this complexity to that of importance sampling. This makes it difficult to assess the practical benefits of the proposed estimator. The paper also suffers from a lack of clarity in its writing. There are instances of undefined notations, grammatical errors, and inconsistencies in equation numbering, which detract from the overall readability and understanding of the paper. For example, the notation θ is used before its definition, and there are duplicate equation numbers. These issues make it difficult to follow the technical arguments and hinder the reader's ability to fully grasp the proposed method. Additionally, the paper's theoretical analysis, while present, has some limitations. The mode-seeking analysis in Section 5.1 is based on the assumption of diagonal covariance matrices, which is a significant limitation. This assumption simplifies the analysis but limits its applicability to more general scenarios with non-diagonal covariance structures. The paper does not adequately discuss the implications of this assumption or provide any theoretical results that do not rely on this simplification. The paper also lacks a comprehensive discussion of the limitations of the proposed method. While the authors mention that ASM may not be suitable for extremely high-dimensional data, they do not provide a detailed analysis of the potential challenges and future research directions. This lack of a balanced perspective makes it difficult to assess the overall impact and applicability of the proposed method. Finally, the experimental results, while promising, are not presented with sufficient detail. The tables lack clear captions and explanations of the evaluation metrics, making it difficult to interpret the results. For example, the meaning of "Sample Quality" and "fe(***)" is not explicitly defined in the table captions. The paper also does not provide a detailed analysis of the experimental results, including a discussion of the strengths and weaknesses of the proposed method in different scenarios. The lack of a clear comparison to other methods and the absence of a discussion of the limitations of the experimental setup further weaken the empirical evaluation. The paper also does not provide a clear explanation of how the MLMC estimator is integrated into the overall ASM algorithm. While Algorithm 2 outlines the steps for the MLMC estimator, the main Algorithm 1 does not explicitly show how the estimator is used to compute the score discrepancy for the parameter update. This lack of clarity makes it difficult to understand the practical implementation of the method. The paper also does not provide a clear explanation of the term $s_{\theta(\tilde{x}_i)}$ in line 5 of Algorithm 1, nor does it clearly state where the imputed samples $\tilde{x}_i$ used in Algorithm 2 are obtained. These omissions further contribute to the lack of clarity in the paper. Based on these observations, I have a high confidence in the validity of these weaknesses.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should provide a more detailed discussion of the relationship between ASM and existing adversarial learning and score matching methods. This should include a clear explanation of how ASM differs from other adversarial score matching techniques, particularly in terms of the specific optimization objectives, the role of the discriminator (or adversary), and the convergence properties. A more thorough comparison with methods like those proposed by Li et al. (2019b) and Binkowski et al. (2023) is needed, highlighting the specific advantages and disadvantages of each approach. Second, the authors should

provide a more detailed explanation of the MLMC estimator, including a clear derivation of its computational complexity and a comparison to the complexity of importance sampling. This should include a discussion of the variance reduction techniques employed and how these techniques are specifically applied to the score discrepancy estimation. The authors should also clarify how the different levels in the MLMC estimator are constructed and how the sample sizes are determined at each level. Third, the authors should carefully revise the paper to address the numerous grammatical errors, inconsistencies in equation numbering, and undefined notations. This should include a thorough proofreading of the paper and a consistent use of notation throughout the manuscript. The authors should also provide clear definitions for all notations before their first use. Fourth, the authors should extend their theoretical analysis to more general scenarios with non-diagonal covariance structures. This could involve exploring alternative assumptions or using more advanced techniques that do not rely on the diagonal covariance assumption. The authors should also discuss the limitations of their current analysis and the potential impact of these limitations on the applicability of their method. Fifth, the authors should provide a more comprehensive discussion of the limitations of the proposed method, including a detailed analysis of the potential challenges in extremely high-dimensional data and a discussion of future research directions. This should include a discussion of the computational cost of the method and how it scales with the dimensionality of the data. Sixth, the authors should provide more detailed explanations of the evaluation metrics used in the experimental results. This should include clear definitions of terms like "Sample Quality" and "fe(***)" in the table captions. The authors should also provide a more detailed analysis of the experimental results, including a discussion of the strengths and weaknesses of the proposed method in different scenarios. The authors should also consider including additional experiments on more diverse datasets and comparing their method to a wider range of baselines. Seventh, the authors should provide a more explicit explanation of how the MLMC estimator is integrated into the overall ASM algorithm. This should include a clear description of how the estimator is used to compute the score discrepancy for the parameter update. The authors should also provide a clear explanation of the term $s_{\theta^t}(\tilde{x}_i)$ in line 5 of Algorithm 1 and clearly state where the imputed samples $\tilde{x}_i$ used in Algorithm 2 are obtained. Finally, the authors should consider adding a dedicated "Limitations" section to the paper to provide a more balanced perspective on the applicability of the proposed method. These changes would significantly improve the clarity, rigor, and overall impact of the paper.

 Questions

I have several questions that arise from my analysis of this paper. First, could the authors elaborate on the specific scenarios where ASM is expected to outperform existing methods, and are there any known limitations or scenarios where ASM might underperform? This would provide a more balanced perspective on the applicability of the proposed method. Second, could the authors provide a more detailed explanation of the specific mechanisms by which the MLMC estimator achieves variance reduction in the context of score discrepancy estimation? This would help to clarify the advantages of the proposed estimator. Third, could the authors provide a more detailed analysis of the computational complexity of the MLMC estimator and compare it to the complexity of importance sampling? This would help to quantify the practical benefits of the proposed estimator. Fourth, could the authors clarify the relationship between the proposed method and existing adversarial training techniques, particularly in terms of the specific optimization objectives, the role of the discriminator (or adversary), and the convergence properties? This would help to better position the contribution of the paper within the broader field of adversarial learning. Fifth, could the authors provide a more detailed explanation of how the missing data mechanism is handled within the score discrepancy calculation? This would help to clarify the applicability of the method to different missing data scenarios. Sixth, could the authors provide a more detailed explanation of the experimental setup, including the specific datasets used, the evaluation metrics, and the hyperparameter settings? This would help to improve the reproducibility of the results. Finally, could the authors clarify the meaning of "n/a" in Table 3 and provide a more detailed explanation of why the method failed in this specific case? These questions target key uncertainties and methodological choices that I believe are critical for a thorough understanding of the paper.

 Scores

Soundness: 2.25

Presentation: 2.0

Contribution: 2.0

Rating: 4.0

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights